Dual-Operator Polishing with Rank Fusion for Reachability Ranking in Uncertain Graphs

Experimental evidence for the W1 algorithm: selection, freeze and confirmation

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Abstract

W1 is a deterministic, training-free heuristic that ranks the vertices of an uncertain graph by their expected reachable mass. It runs a normalised diffusion with the calibrated escape-probability damping rule to its fixed point, polishes the result with three raw propagation steps on a second operator, and fuses the ranks of the two phases. This report documents how W1 was chosen and how it performs. It was selected among 16 800 configurations of a staged search on a development panel of 180 graphs, frozen with a recorded SHA-256 digest, and then evaluated on seven held-out panels with graphs of up to 10,000 vertices, on which it was never tuned. Its mean Spearman correlation with the ground truth is the highest of all deployable comparators on six of the seven panels; on the panel of the largest graphs, two-iteration message passing has a higher mean, without a significant difference. The differences to IRIE and to the calibrated damping rules are significant after Holm correction on both primary panels. W1 also matches or exceeds the per-instance best constant damping, an upper bound of the constant-damping family, on graphs of up to about 150 vertices, but falls below it on graphs of 900 vertices and more; the loss is concentrated in the regime of high edge probabilities, and taking more propagation steps does not remove it. On the 100-instance hold-out of the earlier escape-probability study it ranks second of 22 methods, after Monte Carlo with 200 worlds. Every number of this report is regenerated by the scripts of the accompanying repository, which reproduce the stored results of the source study to within 10^{-16}.

Contents

1 Introduction **2**

2 Materials and methods **2**
2.1 Panels . . . 2
2.2 Ground truth and metrics . . . 3
2.3 Comparators . . . 3
2.4 Statistical procedure . . . 3

3 Selection on the development panel **3**
3.1 Search . . . 3
3.2 What the search found . . . 4
3.3 Freeze . . . 4

4 Confirmation on held-out panels **5**
4.1 Main results . . . 5
4.2 Graph size and the oracle . . . 6
4.3 Comparison with the paper's configuration . . . 7

5 External hold-out of the earlier study **7**

6 Exploratory analyses **8**
6.1 Cost against Monte Carlo 8
6.2 Sensitivity to the two normalisation conventions 9
6.3 Number of Phase-B steps 10
6.4 Edge-probability regimes 10

7 Protocol deviations and limitations **10**

8 Reproduction **11**

A Confirmatory contrasts of W3 **12**

B Secondary metrics **13**

1 Introduction

Let $G = (V, E)$ be an undirected graph whose edges are present independently with probabilities p_{ij}, and let $x > 0$, $\sum_i x_i = 1$, be a prior over the vertices. The expected reachable mass of vertex i is $y_i^* = \mathbb{E}\big[\sum_{v \in C(i)} x_v\big]$, where $C(i)$ is the connected component of i in a sampled world; for uniform x it is the single-seed influence spread of the independent cascade model. The task is to rank the vertices by y^*.

W1 belongs to the dual-operator polishing family introduced in the accompanying paper. Its specification is given in a separate document (`docs/specification/W1_specification_en.pdf`); in brief, Phase A runs the lazy normalised diffusion on the escape-scaled pull operator $EW = \mathrm{diag}(P_{\mathrm{esc}})W$ with the calibrated scalar damping $\alpha_A = \mathrm{clip}(0.502 + 0.558\,\bar{P}_{\mathrm{esc}}, 0.4, 1)$ of the earlier escape-probability study to its fixed point; its output is embedded in the excess scale with $\gamma = 16$; Phase B runs $K = 3$ raw steps $b \leftarrow 0.3x + 0.7\,EP\,b$ with $EP = \mathrm{diag}(P_{\mathrm{esc}})P$; and the output is the rank fusion $0.3\,r(a) + 0.7\,r(b)$. The configuration of the paper itself, Phase B on P with damping 0.9, five steps, $\gamma = 1$ and equal weights, is reported as CTRL.

This report answers three questions. How was W1 chosen, and could the choice have been influenced by the data on which it is evaluated (Section 3)? How does it compare with the alternatives on data it has never seen (Sections 4 and 5)? And where are its limits (Sections 6 and 7)? Section 8 describes how every result is regenerated.

2 Materials and methods

2.1 Panels

Table 1 lists the eight panels, 984 instances in total. The synthetic families are Erdős–Rényi, Barabási–Albert, Watts–Strogatz, square and rectangular grids, stochastic block models, core–periphery graphs, random trees and barbells; the edge-probability regimes draw p_{ij} uniformly from R1 $[0.01, 0.06]$, R2 $[0.06, 0.18]$, R3 $[0.18, 0.40]$, R4 $[0.40, 0.70]$ and R5 $[0.70, 0.95]$. The development panel was used for the selection only. The panels `test` and `prior` were held out from the selection of the accompanying paper and from the selection of W1. The panels `test2`, `scale2`, `prior2`, `real2` and `big` were generated with fresh seeds for the confirmation of W1; `test2` and `scale2` were designated primary before the freeze.

Table 1: Panels. "Ceiling" is the mean Spearman correlation between the two independent ground-truth samples, a measure of the noise of the reference.

Panel	Role	Instances	Vertices	Content	Ceiling
`dev`	selection	180	100	9 families, R1–R5, uniform/Dirichlet(1)	0.956
`test2`	primary confirmation	180	100	9 families, R1–R5, uniform/Dirichlet(1)	0.960
`scale2`	primary confirmation	90	900	9 families, R1–R5, uniform/Dirichlet(1)	0.951
`prior2`	secondary confirmation	150	144–150	5 families, R2–R4, 10 priors	0.978
`real2`	secondary confirmation	24	32–77	3 classic networks, 4 probability schemes	0.989
`big`	secondary confirmation	30	2000–10000	5 families, R2 and R4, uniform	0.970
`test`	earlier held-out	180	100	9 families, R1–R5, uniform/Dirichlet(1)	0.958
`prior`	earlier held-out	150	144–150	5 families, R2–R4, 10 priors	0.978

2.2 Ground truth and metrics

Every instance carries two independent Monte-Carlo estimates of y^* over 2,000 sampled worlds each: y^A for evaluation and y^B for fitting the oracles. The primary metric is the Spearman correlation ρ between a method's scores and y^A, with scores equal to within a relative 10^{-12} treated as ties; Kendall's τ_b, the overlap of the top 10% and NDCG at 10% are secondary. The ceiling column of Table 1 shows that the reference itself is noisy at the level of a few thousandths: differences of that order between methods are within the resolution of the ground truth.

2.3 Comparators

IRIE influence ranking by iterative estimation (Jung, Heo and Chen, ICDM 2012): $y = x + aPy$ with $a = \min(0.7, 0.95/\rho(P))$.

EW rule the escape-scaled pull diffusion with the calibrated damping rule of the earlier study; this is Phase A of W1 on its own.

WT rule personalised PageRank on the push operator $W^\top$, with its own calibrated rule

$$\alpha = \mathrm{clip}(0.241 + 0.859\,\bar{P}_{\mathrm{esc}},\, 0.4,\, 1).$$

MP2 two iterations of the message passing of Karrer, Newman and Zdeborová.

α=0.85 the escape-scaled pull diffusion with the conventional damping 0.85.

oracle for each instance, the constant damping of the escape-scaled pull diffusion (grid of 40 values) whose ranking agrees best with y^B. It uses ground-truth information and cannot be deployed; it is an upper bound of what any rule that predicts one constant damping per graph can achieve.

CTRL the dual-operator configuration of the accompanying paper (added after the fact, Section 4.3).

2.4 Statistical procedure

For each held-out panel, the frozen confirmatory family consists of the twelve contrasts {W1, W3} × {IRIE, EW rule, WT rule, MP2, α=0.85, oracle}. Each contrast is a two-sided Wilcoxon signed-rank test on the per-instance differences of ρ, with exact zero differences dropped; the twelve p-values of a panel are adjusted by Holm's method. The 95% intervals of the mean differences are percentile bootstrap intervals with 20,000 instance-level resamples. W3 is the second frozen configuration; its contrasts are listed in Appendix A.

3 Selection on the development panel

3.1 Search

The search evaluated configurations of the form "Phase A, excess embedding with γ, K raw Phase-B steps on an operator with a damping law, rank fusion with weight β" on all 180 development instances, in four stages.

S0 an audit reproducing the published values of CTRL ($\bar{\rho} = 0.9037$), the EW rule, the WT rule and IRIE.

Comparators 19 methods, among them the oracle (0.9105), IRIE (0.8800) and the EW rule (0.8571).

S2 a full grid of 16 800 configurations: six Phase-B operators (P, EP, the log-weight matrix $P_{\log}$, the row-normalised P, W, EW), 20 damping laws (14 constants from 0.3 to 1 and six multiples of $1/\rho$ of the operator), 14 step counts from 1 to 50, $\beta \in \{0.3, \ldots, 0.7\}$ and $\gamma \in \{1, 4\}$. Best: 0.9116.

S3 a refinement of the 50 best S2 configurations over $\beta \in \{0, 0.05, \ldots, 1\}$ and $\gamma \in \{0.25, \ldots, 16\}$, together with nine variable damping laws (the two calibrated rules, the per-node law $\alpha_i = gS_i/(x_i + gS_i)$ with $S = Ty$, its mean-field version, and the anchored proportional law of the paper): 15 750 evaluations per instance, of which 3 465 are distinct configurations. Best: 0.9133.

S4 two alternating schedules of Phase A and Phase B blocks; best 0.9029.

3.2 What the search found

Table 2 lists the ten best configurations. The optimum is a plateau rather than a peak: 10 configurations lie within 0.001 of the best, 24 within 0.002 and 229 within 0.005. All configurations within 0.002 take exactly three Phase-B steps on one of the raw operators EP, P or $P_{\log}$, with a damping between 0.5 and 0.8 and β between 0.6 and 0.85. The step count is the most consequential parameter (Table 3): the best S2 mean rises from one to three steps and decreases monotonically beyond. Phase B on a normalised operator (W, EW, row-normalised P) is markedly worse (best 0.8359 to 0.8622, against 0.9116 for $P_{\log}$).

No variable damping law reached the plateau. At $\gamma = 4$, the value at which the laws were examined, the best constant damping reaches 0.9116; the best per-node law reaches 0.8916, its mean-field version 0.9083 and the anchored proportional law 0.9056. The search therefore confirms, from the side of the polishing step, the conclusion of the paper that per-node proportional damping does not improve on a scalar damping.

Table 2: The ten best configurations of the development search (all with Phase A as specified).

Rank	Operator	α_B	K	γ	β	Development $\bar{\rho}$
1 (W1)	EP	0.7	3	16	0.7	0.9133
2 (W3)	EP	0.7	3	16	0.75	0.9132
3	EP	0.75	3	16	0.7	0.9130
4	EP	0.75	3	16	0.75	0.9128
5	Plog	0.5	3	16	0.75	0.9127
6	EP	0.7	3	16	0.8	0.9126
7	Plog	0.5	3	16	0.7	0.9126
8	EP	0.7	3	16	0.65	0.9126
9	EP	0.75	3	16	0.65	0.9124
10	Plog	0.5	3	16	0.8	0.9123

Table 3: Best mean development ρ of stage S2 for each number of Phase-B steps K.

K	1	2	3	4	5	6	7	8	10	12	15	20	30	50
best $\bar{\rho}$	0.894	0.906	0.912	0.909	0.909	0.908	0.907	0.906	0.905	0.902	0.900	0.897	0.893	0.889

3.3 Freeze

The selection file `selected_mixing.json` was written at 2026-10-31 17:03:41 UTC. It names W1, with development mean 0.9133, and W3, with 0.9132, identical to W1 except for $\beta = 0.75$. The file and its SHA-256 digest, which begins with `13066bc4f4450904`, were committed at once. The first confirmation results were committed 26 minutes later; the order is recorded in the history of the source repository (`evidence/original/PROVENANCE.md`). The best alternating schedule (0.9029) did not exceed W1, so no schedule was frozen.

The development data cannot distinguish W1 from its neighbours on the plateau. In 2,000 instance-level bootstrap resamples of the development panel, W1 has the highest mean in a fraction 0.257 of the resamples and W3 in 0.174, while 55 different configurations are best at least once (the source study reported 0.274 and 0.184 with a different seed). The confirmation results below should therefore be read as results for the plateau as much as for W1 itself.

4 Confirmation on held-out panels

4.1 Main results

Table 4 gives the mean ρ of every method on every held-out panel. Among the deployable methods, W1 or W3 has the highest mean on six of the seven panels; on `big`, MP2 is higher (0.9351 against 0.9096). W1 exceeds the EW rule, whose damping it uses in Phase A, by $+0.0660$ on `test2` and $+0.0600$ on `scale2`, and IRIE by $+0.0223$ and $+0.0348$.

Table 4: Mean Spearman ρ on the held-out panels. Bold: best deployable method of the panel (the oracle is excluded). Instances per panel as in Table 1.

Method	`test2`	`scale2`	`prior2`	`real2`	`big`	`test`	`prior`
W1	0.9064	0.8650	**0.9183**	0.9625	0.9096	0.9103	**0.9049**
W3	**0.9072**	**0.8651**	0.9162	**0.9638**	0.9099	**0.9108**	0.9033
CTRL	0.8950	0.8611	0.9158	0.9463	0.9008	0.9006	0.9016
oracle	0.9009	0.8775	0.9109	0.9428	0.9570	0.9044	0.9002
IRIE	0.8841	0.8302	0.8944	0.9398	0.8850	0.8960	0.8906
EW rule	0.8404	0.8050	0.8763	0.9013	0.8451	0.8442	0.8568
WT rule	0.8153	0.7474	0.8599	0.9124	0.8005	0.8113	0.8554
MP2	0.7491	0.7482	0.7859	0.9060	**0.9351**	0.7539	0.7789
α=0.85	0.7785	0.7471	0.6930	0.8148	0.8971	0.7806	0.6927

Table 5 gives the confirmatory contrasts of W1. After Holm correction within each panel:

- W1 is significantly better than the EW rule on all seven panels, and than the WT rule and α=0.85 on all panels except `big`;
- W1 is significantly better than IRIE on the two primary panels (`test2`: $p_{\text{Holm}} = 8.5 \cdot 10^{-4}$; `scale2`: 0.039) and on `test`; on `prior2`, `real2`, `big` and `prior` the mean differences are positive but not significant after correction;
- against MP2 the differences are significant on `test2`, `prior2`, `test` and `prior`, not significant on `scale2` and `real2`, and negative but not significant on `big` (-0.0255);
- against the oracle, W1 is significantly better on `test` ($+0.0059$), not significantly different on `test2`, `prior2`, `real2` and `prior`, and significantly worse on `scale2` (-0.0126) and `big` (-0.0474).

The preregistered non-inferiority margin of 0.005 against the oracle on the primary panels is therefore met on `test2` and not on `scale2`.

Table 5: Confirmatory contrasts of W1: mean difference of ρ with 95% bootstrap interval, wins/losses over instances, Wilcoxon p and Holm-adjusted p (12 contrasts per panel). *: $p_{\text{Holm}} < 0.05$.

Panel	Comparator	$\Delta\bar{\rho}$	95% CI	W/L	p	p_{Holm}
`test2`	IRIE	$+0.0223^*$	[+0.0120, +0.0331]	95/85	$2.8 \cdot 10^{-4}$	$8.5 \cdot 10^{-4}$
	EW rule	$+0.0660^*$	[+0.0505, +0.0823]	119/61	$3.6 \cdot 10^{-13}$	$3.6 \cdot 10^{-12}$
	WT rule	$+0.0911^*$	[+0.0684, +0.1154]	128/52	$2.6 \cdot 10^{-12}$	$2.1 \cdot 10^{-11}$
	MP2	$+0.1573^*$	[+0.1130, +0.2046]	100/80	$3.9 \cdot 10^{-6}$	$2.0 \cdot 10^{-5}$
	α=0.85	$+0.1279^*$	[+0.0977, +0.1609]	159/21	$1.9 \cdot 10^{-22}$	$2.3 \cdot 10^{-21}$

Table 5: Confirmatory contrasts of W1: mean difference of ρ with 95% bootstrap interval, wins/losses over instances, Wilcoxon p and Holm-adjusted p (12 contrasts per panel). *: $p_{\text{Holm}} < 0.05$.

Panel	Comparator	$\Delta\bar{\rho}$	95% CI	W/L	p	p_{Holm}
	oracle	$+0.0055$	$[-0.0067, +0.0183]$	74/106	0.071	0.141
`scale2`	IRIE	$+0.0348^*$	$[+0.0085, +0.0626]$	50/40	0.013	0.039
	EW rule	$+0.0600^*$	$[+0.0389, +0.0822]$	56/34	$3.3 \cdot 10^{-5}$	$2.6 \cdot 10^{-4}$
	WT rule	$+0.1176^*$	$[+0.0745, +0.1620]$	64/26	$3.3 \cdot 10^{-6}$	$3.3 \cdot 10^{-5}$
	MP2	$+0.1167$	$[+0.0515, +0.1866]$	46/44	0.071	0.142
	α=0.85	$+0.1178^*$	$[+0.0757, +0.1645]$	73/17	$4.2 \cdot 10^{-8}$	$4.9 \cdot 10^{-7}$
	oracle	-0.0126^*	$[-0.0338, +0.0094]$	28/62	0.001	0.009
`prior2`	IRIE	$+0.0240$	$[+0.0112, +0.0380]$	80/70	0.029	0.117
	EW rule	$+0.0420^*$	$[+0.0300, +0.0545]$	88/62	$8.5 \cdot 10^{-8}$	$8.5 \cdot 10^{-7}$
	WT rule	$+0.0584^*$	$[+0.0370, +0.0803]$	100/50	$2.1 \cdot 10^{-6}$	$1.6 \cdot 10^{-5}$
	MP2	$+0.1325^*$	$[+0.0892, +0.1768]$	75/75	$4.9 \cdot 10^{-4}$	0.003
	α=0.85	$+0.2253^*$	$[+0.1833, +0.2692]$	114/36	$1.3 \cdot 10^{-17}$	$1.6 \cdot 10^{-16}$
	oracle	$+0.0074$	$[-0.0037, +0.0188]$	64/86	0.925	1.000
`real2`	IRIE	$+0.0226$	$[+0.0029, +0.0440]$	15/9	0.049	0.245
	EW rule	$+0.0611^*$	$[+0.0360, +0.0891]$	22/2	$3.0 \cdot 10^{-6}$	$3.0 \cdot 10^{-5}$
	WT rule	$+0.0500^*$	$[+0.0203, +0.0805]$	19/5	0.007	0.046
	MP2	$+0.0564$	$[-0.0005, +0.1254]$	10/13	0.917	1.000
	α=0.85	$+0.1476^*$	$[+0.0672, +0.2456]$	22/2	$1.2 \cdot 10^{-6}$	$1.4 \cdot 10^{-5}$
	oracle	$+0.0197$	$[-0.0053, +0.0481]$	15/9	0.169	0.506
`big`	IRIE	$+0.0247$	$[-0.0146, +0.0593]$	20/10	0.026	0.099
	EW rule	$+0.0645^*$	$[+0.0241, +0.1053]$	24/6	$4.6 \cdot 10^{-4}$	0.006
	WT rule	$+0.1091$	$[+0.0327, +0.1927]$	17/13	0.016	0.096
	MP2	-0.0255	$[-0.0647, +0.0123]$	17/13	0.416	0.832
	α=0.85	$+0.0125$	$[-0.0290, +0.0472]$	26/4	0.014	0.096
	oracle	-0.0474^*	$[-0.0842, -0.0173]$	9/21	0.003	0.026
`test`	IRIE	$+0.0143^*$	$[-0.0019, +0.0282]$	96/84	$5.6 \cdot 10^{-4}$	0.002
	EW rule	$+0.0661^*$	$[+0.0489, +0.0845]$	119/61	$2.9 \cdot 10^{-11}$	$2.3 \cdot 10^{-10}$
	WT rule	$+0.0990^*$	$[+0.0746, +0.1237]$	129/51	$1.4 \cdot 10^{-12}$	$1.2 \cdot 10^{-11}$
	MP2	$+0.1564^*$	$[+0.1121, +0.2034]$	106/74	$1.2 \cdot 10^{-6}$	$5.9 \cdot 10^{-6}$
	α=0.85	$+0.1297^*$	$[+0.0971, +0.1641]$	151/29	$2.1 \cdot 10^{-20}$	$2.5 \cdot 10^{-19}$
	oracle	$+0.0059^*$	$[-0.0084, +0.0206]$	73/107	0.020	0.040
`prior`	IRIE	$+0.0142$	$[+0.0009, +0.0284]$	73/77	0.933	1.000
	EW rule	$+0.0481^*$	$[+0.0328, +0.0632]$	87/63	$1.0 \cdot 10^{-6}$	$1.0 \cdot 10^{-5}$
	WT rule	$+0.0495^*$	$[+0.0265, +0.0724]$	90/60	0.001	0.007
	MP2	$+0.1260^*$	$[+0.0823, +0.1708]$	77/73	$1.6 \cdot 10^{-4}$	0.001
	α=0.85	$+0.2121^*$	$[+0.1711, +0.2543]$	112/38	$1.1 \cdot 10^{-15}$	$1.3 \cdot 10^{-14}$
	oracle	$+0.0046$	$[-0.0089, +0.0182]$	52/98	0.032	0.097

The secondary metrics agree with ρ (Appendix B).

4.2 Graph size and the oracle

Table 6 orders the panels by graph size. The difference between W1 and the oracle decreases with the size of the graphs: W1 exceeds the per-instance best constant damping on the panels with at most about 150 vertices and falls below it on the panels with 900 vertices and more. The margin over IRIE does not shrink in the same way. Sections 6.3 and 6.4 show that the loss is not due to too few propagation steps and that it is concentrated at high edge probabilities.

Table 6: Mean ρ of W1, the oracle and IRIE by panel, ordered by graph size.

Panel	Vertices	W1	oracle	W1 − oracle	IRIE	W1 − IRIE
`real2`	32–77	0.9625	0.9428	+0.0197	0.9398	+0.0226
`test2`	100	0.9064	0.9009	+0.0055	0.8841	+0.0223
`test`	100	0.9103	0.9044	+0.0059	0.8960	+0.0143
`prior2`	144–150	0.9183	0.9109	+0.0074	0.8944	+0.0240
`prior`	144–150	0.9049	0.9002	+0.0046	0.8906	+0.0142
`scale2`	900	0.8650	0.8775	−0.0126	0.8302	+0.0348
`big`	2000–10000	0.9096	0.9570	−0.0474	0.8850	+0.0247

4.3 Comparison with the paper's configuration

CTRL belonged to the declared comparator family but was not evaluated in the stored confirmation runs; it is evaluated here with the same code, after the fact, and reported separately with a Holm correction across the seven panels (Table 7). W1 has the higher mean on every panel; the difference is significant on `test2`, `real2` and `test`.

Table 7: W1 against CTRL (post hoc). p_{Holm}: Holm adjustment across the seven panels.

Panel	W1	CTRL	$\Delta\bar{\rho}$	W/L	p_{Holm}
`test2`	0.9064	0.8950	+0.0114	107/73	$9.0 \cdot 10^{-4}$
`scale2`	0.8650	0.8611	+0.0038	48/42	0.571
`prior2`	0.9183	0.9158	+0.0026	75/75	0.673
`real2`	0.9625	0.9463	+0.0161	20/4	$1.4 \cdot 10^{-4}$
`big`	0.9096	0.9008	+0.0089	19/11	0.561
`test`	0.9103	0.9006	+0.0097	107/73	0.011
`prior`	0.9049	0.9016	+0.0033	76/74	0.673

5 External hold-out of the earlier study

The earlier escape-probability study evaluated 22 methods (including those added here) on 100 instances of 100 vertices (Erdős–Rényi, Barabási–Albert, Watts–Strogatz and grid; five probability ranges; Dirichlet priors), with a ground truth of 3,000 worlds per instance. Its instances and ground truth are rebuilt with its own code, vendored unmodified. W1 ranks 2 of 22 (Table 8), with $\bar{\rho} = 0.9310$, after Monte Carlo with 200 worlds (0.9335) and ahead of IRIE (0.9142), the per-instance best constant damping (0.9108), the calibrated escape-scaled rule of that study (0.8940) and PMIA (0.8472). By topology (Table 9), W1 is ahead of IRIE on Erdős–Rényi, Barabási–Albert and Watts–Strogatz graphs and behind it on grids.

The values of the earlier study are its stored results, computed with scipy's Spearman correlation; W1, W3 and CTRL are scored with the tie-tolerant Spearman of this project, as in the source study. Rank-fused scores contain exact ties that rounding can split; with scipy's Spearman the values of W1 change by at most $4 \cdot 10^{-4}$ per instance.

Table 8: The 100-instance hold-out of the earlier escape-probability study: mean ρ of every method.

#	Method	$\bar{\rho}$
1	Monte Carlo, 200 worlds	0.9335
2	**W1 (this work)**	0.9310
3	CTRL (paper configuration)	0.9296
4	W3 (this work)	0.9289
5	IRIE	0.9142
6	oracle (best constant damping)	0.9108
7	escape-scaled pull, calibrated rule	0.8940
8	PPR push, calibrated rule	0.8919
9	PPR pull $\times P_{\text{esc}}$	0.8640
10	Monte Carlo, 50 worlds	0.8506
11	PMIA	0.8472
12	PPR push, $\alpha = 0.85$	0.7253
13	Monte Carlo, 10 worlds	0.6770
14	$\sum_j p_{ij}$	0.6234
15	degree discount	0.6234
16	P_{esc}	0.6216
17	RIS, 50 samples	0.6188
18	escape-scaled pull, $\alpha = 1$	0.6002
19	PPR pull, calibrated rule	0.4385
20	IRIE, row-normalised	0.4214
21	PPR pull, $\alpha = 0.85$	0.4048
22	prior only	0.3893

Table 9: External hold-out by topology (25 instances each).

Topology	W1	MC-200	IRIE	escape-scaled rule
Erdős–Rényi	0.960	0.953	0.954	0.898
Barabási–Albert	0.944	0.965	0.880	0.917
Watts–Strogatz	0.864	0.841	0.851	0.818
grid	0.957	0.975	0.971	0.943

6 Exploratory analyses

The analyses of this section were not preregistered and use the confirmation panels after the fact. They describe the frozen W1; none of them was used to change it.

6.1 Cost against Monte Carlo

Monte-Carlo estimation with k sampled worlds is the direct alternative to any heuristic. Table 10 compares W1 with Monte Carlo on identical instances, single-threaded on one machine (W1 timed end to end, operator construction included; each timing the minimum of three repetitions, one on graphs above 1,000 vertices). On graphs of 100 vertices (`test2`), Monte Carlo with 200 worlds is faster than W1 (median 1.6 against 2.8 ms) and about as accurate (W1 $-$ MC-200: $+0.005$, wins/losses 94/86, $p = 0.080$), and Monte Carlo with 500 worlds is more accurate (-0.022, $p = 0.042$) at 1.3 times the time of W1. On graphs of 900 vertices (`scale2`), W1 is 2.8 times faster than 200 worlds, with a similar accuracy (-0.019, 44/46, $p = 0.833$), and 500 worlds are significantly more accurate (-0.056, $p = 1.6 \cdot 10^{-4}$) at 7.3 times its time. On graphs of 2,000 to 10,000 vertices (`big`), W1 is significantly more accurate than 200 worlds ($+0.033$, 24/6, $p = 0.015$) and 3.3 times faster, and 500 worlds, at 8.3 times its time, are not significantly better ($p = 0.715$). These tests are unadjusted and exploratory.

Table 10: Mean ρ / median time in milliseconds of W1 and of Monte Carlo with k worlds (exploratory; timings are machine-specific).

Panel	Vertices	W1	MC-50	MC-100	MC-200	MC-500
`test2`	100	0.906 / 2.8	0.793 / 0.6	0.860 / 0.9	0.901 / 1.6	0.929 / 3.7
`scale2`	900	0.865 / 5.0	0.792 / 3.7	0.844 / 7.2	0.884 / 14.0	0.921 / 36.3
`big`	2000–10000	0.910 / 23.1	0.731 / 19.9	0.812 / 37.0	0.877 / 77.3	0.932 / 192.1

The comparison depends on the edge probabilities (Table 11, graphs of 900 vertices and more). With low probabilities (R1, R2), W1 is more accurate than 200 worlds and on a par with 500 (R2: 0.963, 0.915 and 0.956). From R3 upwards Monte Carlo is more accurate: with high probabilities a giant component forms in most worlds, the ranking depends on which vertices join it, a global and non-linear effect that a linear diffusion captures less well, and Monte Carlo is the better choice. Phase A did not converge within 20,000 iterations on 20 of the 300 instances (barbells, rectangular and square grids and trees with high edge probabilities, where the calibrated rule returns $\alpha_A = 1$, the boundary case of the specification); these instances account for the long tail of its running times.

Table 11: Graphs of 900 vertices and more (`scale2` and `big`) by edge-probability regime (exploratory).

Edge probabilities	Instances	W1	MC-200	MC-500	W1 ms	MC-200 ms
R1 [0.01, 0.06]	18	0.955	0.914	0.953	3.1	6.4
R2 [0.06, 0.18]	33	0.963	0.915	0.956	5.5	17.3
R3 [0.18, 0.4]	18	0.924	0.933	0.974	6.7	14.7
R4 [0.4, 0.7]	33	0.856	0.900	0.943	24.2	34.4
R5 [0.7, 0.95]	18	0.627	0.707	0.751	16.7	21.8

6.2 Sensitivity to the two normalisation conventions

Table 12 varies the two scale conventions of the specification. Dividing the Phase-B starting point by $1+\gamma$ lowers ρ on every panel: the recursion of Phase B is affine, so the scale of its starting point relative to x is a parameter, and the excess-scale convention fixes it. Without the per-step normalisation, Phase A converges to the resolvent $(1-\alpha_A)(I-\alpha_A EW)^{-1}x$ instead of the Perron vector of the specification, which equals the resolvent with the larger effective damping α_A/λ (λ the Perron root); when $\alpha_A = 1$ (148 of 474 instances) its mass decays geometrically, so that the stopping rule is met because the iterate becomes small rather than because it has stabilised. Phase A on its own is worse without the normalisation on every panel. The effect on W1 is mixed: removing the normalisation lowers ρ on `test2`, `prior2` and `real2`, leaves it unchanged on `scale2` and raises it on `big`. The gain is confined to the high-probability regimes, where α_A is at or near 1 and the normalised Phase A approaches the Perron vector of EW alone (`big`, R4: mean $\alpha_A = 0.998$, W1 0.848, without normalisation 0.871); in the other regimes the normalisation helps. The calibrated rule was fitted with the normalised iteration, so this does not suggest removing it; it marks the behaviour of Phase A at $\alpha_A \approx 1$ as the place where W1 can be improved.

Table 12: Mean ρ of W1 and of two variants (exploratory).

Panel	W1	bridge $/(1+\gamma)$	no Phase-A normalisation	Phase A only
`test2`	0.9064	0.8910	0.9046	0.8404
`scale2`	0.8650	0.8427	0.8652	0.8050
`prior2`	0.9183	0.8774	0.9030	0.8763
`real2`	0.9625	0.9521	0.9611	0.9013
`big`	0.9096	0.9074	0.9204	0.8451

6.3 Number of Phase-B steps

The number of Phase-B steps was selected on graphs of 100 vertices only. Table 13 varies K with every other parameter frozen.

Three steps are best on average on every panel, including the panel of 2,000 to 10,000 vertices ($K = 1$: 0.9091; $K = 3$: 0.9096; $K = 20$: 0.8815). More propagation therefore does not close the gap to the oracle on large graphs. More steps help only in individual cells: on grids of the `big` panel with high probabilities ($K = 3$: 0.737; $K = 20$: 0.763) and at the lowest probabilities of `scale2` (R1, $K = 3$: 0.955; $K = 12$: 0.971); on the other high-probability cells of `big` they hurt (Watts–Strogatz, R4: 0.826 against 0.671).

Table 13: Mean ρ of W1 with K Phase-B steps, other parameters frozen (exploratory). Bold: best K of the panel.

Panel	K=1	K=2	K=3	K=4	K=5	K=6	K=8	K=12	K=20
`test2`	0.889	0.902	**0.906**	0.903	0.901	0.899	0.896	0.893	0.887
`scale2`	0.845	0.860	**0.865**	0.864	0.863	0.861	0.860	0.858	0.855
`prior2`	0.877	0.903	**0.918**	0.917	0.917	0.914	0.912	0.907	0.901
`real2`	0.944	0.949	**0.962**	0.951	0.954	0.946	0.944	0.943	0.943
`big`	0.909	0.909	**0.910**	0.904	0.899	0.896	0.891	0.886	0.882

6.4 Edge-probability regimes

Table 14 breaks the primary panels and `big` down by edge-probability regime.

At the lowest edge probabilities (R1) the leading methods are close to the noise ceiling and W1 is marginally below IRIE, the EW rule and the oracle (`test2`: W1 0.985, IRIE 0.988, oracle 0.989). From R2 upwards W1 is ahead of IRIE and the EW rule on the primary panels, and its margin grows with the edge probabilities (`test2`, R3: W1 0.937, IRIE 0.923, EW rule 0.889; R4: W1 0.891, IRIE 0.839, EW rule 0.799). At high probabilities, however, two-iteration message passing is the strongest deployable method (`test2`, R5: MP2 0.815, W1 0.740; `big`, R4: MP2 0.923, W1 0.848), and on large graphs the oracle is far ahead (`big`, R4: 0.953). The loss of W1 against the oracle on large graphs is therefore a loss in the supercritical regime, in which a giant component forms in most worlds and the calibrated rule sets α_A at or near 1; at low probabilities W1 exceeds the oracle on `big` (R2: 0.972 against 0.961). A selector that hands the high-probability regime to message passing is a natural next step; it was not part of the search.

Table 14: Mean ρ by edge-probability regime (exploratory).

Panel	Regime	n	W1	oracle	IRIE	MP2	EW rule
`test2`	R1	36	0.985	0.989	0.988	0.504	0.989
	R2	36	0.979	0.984	0.975	0.682	0.973
	R3	36	0.937	0.935	0.923	0.840	0.889
	R4	36	0.891	0.892	0.839	0.905	0.799
	R5	36	0.740	0.706	0.696	0.815	0.552
`scale2`	R1	18	0.955	0.978	0.962	0.550	0.943
	R2	18	0.956	0.965	0.939	0.707	0.923
	R3	18	0.924	0.923	0.880	0.836	0.879
	R4	18	0.863	0.885	0.792	0.877	0.787
	R5	18	0.627	0.637	0.578	0.771	0.492
`big`	R2	15	0.972	0.961	0.975	0.948	0.934
	R4	15	0.848	0.953	0.795	0.923	0.756

7 Protocol deviations and limitations

1. **Tie-break of W1.** The declared rule was: among configurations within 0.001 of the best, prefer fewer free parameters, then lower measured cost. All 10 candidates have the same parameter set, and the freeze

took the configuration with the highest mean. Applied to the stored cost estimates, the cost criterion would have chosen the configuration ranked 5 (operator $P_{\log}$, damping 0.5, $\beta = 0.75$; development mean 0.9127), whose median cost (0.728 ms) differs from that of W1 (0.752 ms) by less than the run-to-run variation of such timings.

2. **Rule of W3.** W3 was declared as the configuration with the fewest free parameters within 0.002 of W1; since all candidates have the same parameter set, the rule does not single out a simpler configuration, and the frozen W3 is the second-ranked configuration.

3. **Bootstrap.** The source study did not commit the code of its bootstrap of the top-1 frequencies; they are recomputed here with this repository's seed.

4. **Comparator family.** The declared family named a "best constant damping on the development panel"; the stored confirmation runs used the conventional $\alpha = 0.85$ and the per-instance oracle instead. CTRL, also declared, was missing from the stored runs and is reported here separately (Section 4.3).

5. **Panel `real2`.** Only the 24 instances built on three classic networks exist; the snowball samples of four larger networks in the original specification could not be built, because the source graphs were unavailable. Conclusions about real networks rest on few, small graphs.

6. **Labels of the source statistics files.** The columns `t` and `l` of the source study's statistics files contain losses and ties; the reproduced files name them accordingly.

7. **Boundaries of the search.** $\gamma = 16$ is the largest value of its grid. On the development panel, larger values give 0.9127 ($\gamma = 32$), 0.9102 (64) and 0.9068 (128), against 0.9133 at $\gamma = 16$, so the optimum lies inside the extended range (exploratory). The step count and every other parameter were selected on graphs of 100 vertices; Section 6.3 shows that $K = 3$ remains the best choice on larger graphs.

8. **Spectral radius.** The damping family "$c/\rho(T)$" of stage S2 computed $\rho(T)$ with a symmetric eigensolver also for non-symmetric operators. The frozen configurations do not use this family; it is kept unchanged for reproducibility.

9. **Scope.** All graphs have at most 10,000 vertices, edges are independent, the ground truth is itself a Monte-Carlo estimate, and timings are of a Python implementation on one machine.

8 Reproduction

The repository contains the code, the data and the stored results of the source study (`evidence/original`), and regenerates every result in `evidence/reproduced`:

- all 984 instances are rebuilt from their specifications and are identical bit for bit (`scripts/verify_data.py`); 16 ground-truth samples were recomputed and are identical;
- the digest of the frozen selection matches, W1 is the best development configuration of the stored search, and the frozen development means are recovered exactly (`scripts/verify_freeze.py`);
- 63 900 re-evaluations of 355 search configurations on the development panel agree with the stored values to within 10^{-16} (`scripts/run_search.py`);
- the 6 432 stored confirmation rows, four metrics each, are reproduced to within 10^{-16} (`scripts/run_confirm.py`), and the stored statistics to within 10^{-16} (`scripts/analyze.py`);
- the external hold-out values of W1 and W3 are reproduced exactly (`scripts/run_external_holdout.py`).

The script `reproduce.sh` runs every step; `scripts/make_doc_assets.py` writes every number of this report.

A Confirmatory contrasts of W3

Table 15: Confirmatory contrasts of W3 (as in Table 5).

Panel	Comparator	$\Delta\bar{p}$	95% CI	W/L	p	p_{Holm}
`test2`	IRIE	$+0.0231^{*}$	$[+0.0132, +0.0335]$	96/84	$1.1 \cdot 10^{-4}$	$4.5 \cdot 10^{-4}$
	EW rule	$+0.0668^{*}$	$[+0.0509, +0.0838]$	118/62	$6.6 \cdot 10^{-12}$	$4.6 \cdot 10^{-11}$
	WT rule	$+0.0920^{*}$	$[+0.0694, +0.1165]$	127/53	$1.5 \cdot 10^{-12}$	$1.4 \cdot 10^{-11}$
	MP2	$+0.1582^{*}$	$[+0.1138, +0.2046]$	99/81	$2.6 \cdot 10^{-6}$	$1.5 \cdot 10^{-5}$
	α=0.85	$+0.1287^{*}$	$[+0.0980, +0.1614]$	158/22	$4.5 \cdot 10^{-22}$	$5.0 \cdot 10^{-21}$
	oracle	$+0.0063$	$[-0.0061, +0.0198]$	76/104	0.077	0.141
`scale2`	IRIE	$+0.0349^{*}$	$[+0.0087, +0.0619]$	50/40	0.006	0.025
	EW rule	$+0.0602^{*}$	$[+0.0375, +0.0838]$	55/35	$7.6 \cdot 10^{-5}$	$5.3 \cdot 10^{-4}$
	WT rule	$+0.1177^{*}$	$[+0.0735, +0.1630]$	64/26	$3.5 \cdot 10^{-6}$	$3.3 \cdot 10^{-5}$
	MP2	$+0.1169$	$[+0.0521, +0.1869]$	47/43	0.078	0.142
	α=0.85	$+0.1180^{*}$	$[+0.0749, +0.1644]$	73/17	$4.1 \cdot 10^{-8}$	$4.9 \cdot 10^{-7}$
	oracle	-0.0124^{*}	$[-0.0337, +0.0099]$	28/62	0.002	0.009
`prior2`	IRIE	$+0.0218$	$[+0.0091, +0.0352]$	77/73	0.058	0.175
	EW rule	$+0.0399^{*}$	$[+0.0270, +0.0529]$	86/64	$1.0 \cdot 10^{-6}$	$9.1 \cdot 10^{-6}$
	WT rule	$+0.0563^{*}$	$[+0.0341, +0.0783]$	96/54	$6.7 \cdot 10^{-6}$	$4.7 \cdot 10^{-5}$
	MP2	$+0.1303^{*}$	$[+0.0874, +0.1745]$	75/75	$5.7 \cdot 10^{-4}$	0.003
	α=0.85	$+0.2232^{*}$	$[+0.1819, +0.2658]$	113/37	$5.4 \cdot 10^{-17}$	$5.9 \cdot 10^{-16}$
	oracle	$+0.0052$	$[-0.0065, +0.0172]$	64/86	0.624	1.000
`real2`	IRIE	$+0.0240$	$[+0.0048, +0.0455]$	16/8	0.031	0.189
	EW rule	$+0.0625^{*}$	$[+0.0370, +0.0917]$	22/2	$3.9 \cdot 10^{-6}$	$3.5 \cdot 10^{-5}$
	WT rule	$+0.0514^{*}$	$[+0.0214, +0.0824]$	19/5	0.004	0.035
	MP2	$+0.0578$	$[+0.0020, +0.1251]$	10/13	0.893	1.000
	α=0.85	$+0.1490^{*}$	$[+0.0685, +0.2461]$	22/2	$1.2 \cdot 10^{-6}$	$1.4 \cdot 10^{-5}$
	oracle	$+0.0210$	$[-0.0037, +0.0496]$	16/8	0.121	0.483
`big`	IRIE	$+0.0250$	$[-0.0118, +0.0570]$	20/10	0.011	0.085
	EW rule	$+0.0648^{*}$	$[+0.0214, +0.1088]$	24/6	$7.3 \cdot 10^{-4}$	0.008
	WT rule	$+0.1094$	$[+0.0331, +0.1932]$	17/13	0.025	0.099
	MP2	-0.0252	$[-0.0668, +0.0140]$	17/13	0.440	0.832
	α=0.85	$+0.0128$	$[-0.0314, +0.0494]$	26/4	0.014	0.096
	oracle	-0.0471^{*}	$[-0.0851, -0.0161]$	9/21	0.003	0.026
`test`	IRIE	$+0.0148^{*}$	$[-0.0006, +0.0284]$	98/82	$3.8 \cdot 10^{-4}$	0.002
	EW rule	$+0.0665^{*}$	$[+0.0491, +0.0853]$	114/66	$6.3 \cdot 10^{-10}$	$4.4 \cdot 10^{-9}$
	WT rule	$+0.0994^{*}$	$[+0.0755, +0.1245]$	127/53	$9.2 \cdot 10^{-13}$	$9.2 \cdot 10^{-12}$
	MP2	$+0.1569^{*}$	$[+0.1123, +0.2034]$	106/74	$7.1 \cdot 10^{-7}$	$4.2 \cdot 10^{-6}$
	α=0.85	$+0.1302^{*}$	$[+0.0978, +0.1648]$	151/29	$9.3 \cdot 10^{-20}$	$1.0 \cdot 10^{-18}$
	oracle	$+0.0064^{*}$	$[-0.0084, +0.0216]$	74/106	0.023	0.040
`prior`	IRIE	$+0.0127$	$[-0.0007, +0.0264]$	73/77	0.887	1.000
	EW rule	$+0.0465^{*}$	$[+0.0304, +0.0629]$	84/66	$9.6 \cdot 10^{-6}$	$8.7 \cdot 10^{-5}$
	WT rule	$+0.0479^{*}$	$[+0.0242, +0.0720]$	89/61	0.002	0.009
	MP2	$+0.1244^{*}$	$[+0.0809, +0.1693]$	77/73	$2.5 \cdot 10^{-4}$	0.002
	α=0.85	$+0.2106^{*}$	$[+0.1704, +0.2531]$	110/40	$8.8 \cdot 10^{-15}$	$9.7 \cdot 10^{-14}$
	oracle	$+0.0030$	$[-0.0110, +0.0175]$	52/98	0.024	0.097

B Secondary metrics

Table 16: Mean Kendall τ_b.

Method	test2	scale2	prior2	real2	big	test	prior
W1	0.7964	**0.7465**	**0.7857**	0.8681	0.7738	0.7997	**0.7722**
W3	**0.7971**	0.7462	0.7833	**0.8722**	0.7756	**0.8001**	0.7702
CTRL	0.7876	0.7456	0.7852	0.8398	0.7591	0.7934	0.7718
oracle	0.8027	0.7771	0.7897	0.8411	0.8384	0.8077	0.7830
IRIE	0.7699	0.7058	0.7644	0.8358	0.7481	0.7810	0.7613
EW rule	0.7331	0.6959	0.7398	0.7781	0.7023	0.7373	0.7265
WT rule	0.6699	0.6032	0.7116	0.7914	0.6496	0.6658	0.7095
MP2	0.6316	0.6335	0.6727	0.8177	**0.8211**	0.6357	0.6661
α=0.85	0.6570	0.6243	0.5736	0.6981	0.7523	0.6588	0.5747

Table 17: Mean overlap of the top 10%.

Method	test2	scale2	prior2	real2	big	test	prior
W1	0.6889	0.6260	0.7391	0.6562	0.6911	0.6867	0.7227
W3	0.6972	**0.6284**	0.7418	0.6823	0.6986	0.6883	0.7298
CTRL	0.6444	0.6033	0.6956	0.6042	0.6523	0.6556	0.7071
oracle	0.6472	0.6090	0.6787	0.6042	0.7437	0.6578	0.6787
IRIE	**0.7094**	0.6281	**0.7573**	**0.7448**	0.7036	**0.7100**	**0.7702**
EW rule	0.5633	0.5379	0.6004	0.4740	0.5475	0.5833	0.6129
WT rule	0.5761	0.5401	0.7156	0.6562	0.6315	0.5789	0.7240
MP2	0.5333	0.5373	0.5996	0.5990	**0.7725**	0.5144	0.5871
α=0.85	0.5317	0.5007	0.5200	0.4375	0.5993	0.5428	0.5213

Table 18: Mean NDCG at 10%.

Method	test2	scale2	prior2	real2	big	test	prior
W1	0.9818	0.9657	0.9769	0.9773	0.9803	0.9816	0.9716
W3	0.9837	0.9664	0.9773	0.9779	0.9806	0.9819	0.9726
CTRL	0.9752	0.9639	0.9704	0.9692	0.9783	0.9784	0.9634
oracle	0.9788	0.9680	0.9716	0.9788	0.9736	0.9806	0.9679
IRIE	**0.9840**	**0.9783**	**0.9810**	**0.9878**	0.9715	**0.9869**	**0.9814**
EW rule	0.9555	0.9455	0.9496	0.9535	0.9603	0.9630	0.9405
WT rule	0.9196	0.8878	0.9600	0.9540	0.9383	0.9187	0.9612
MP2	0.8615	0.8786	0.8019	0.9247	**0.9819**	0.8634	0.7960
α=0.85	0.9436	0.9359	0.9150	0.9359	0.9647	0.9447	0.9113